

Note: this should be a short and simple review, so PLEASE still study your notes and do practice problems. also PLEASE go over work-energy, it finds its way in many momentum problems

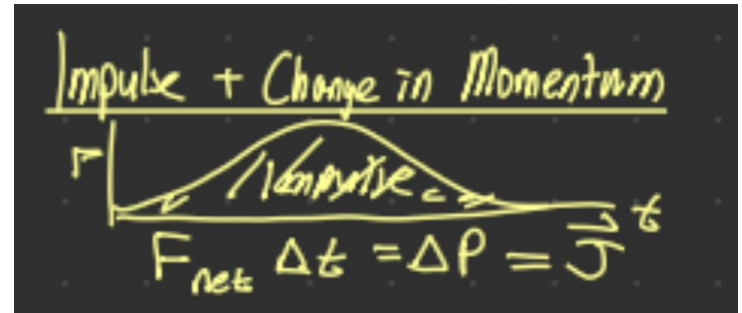
Unit 4: Momentum & Impulse

Momentum & Impulse: Basic Definitions

- Momentum (kg*m/s)
 - Quantity of motion for an object → denoted by $p = mv$,
 - m is the mass of the object and v is the velocity
 - Because v is a vector, p is also a vector quantity
- Impulse
 - Change in momentum is called impulse → denoted by $J = \Delta p = m\Delta v$
- Area under the F vs Δt curve gives us the momentum over some time interval

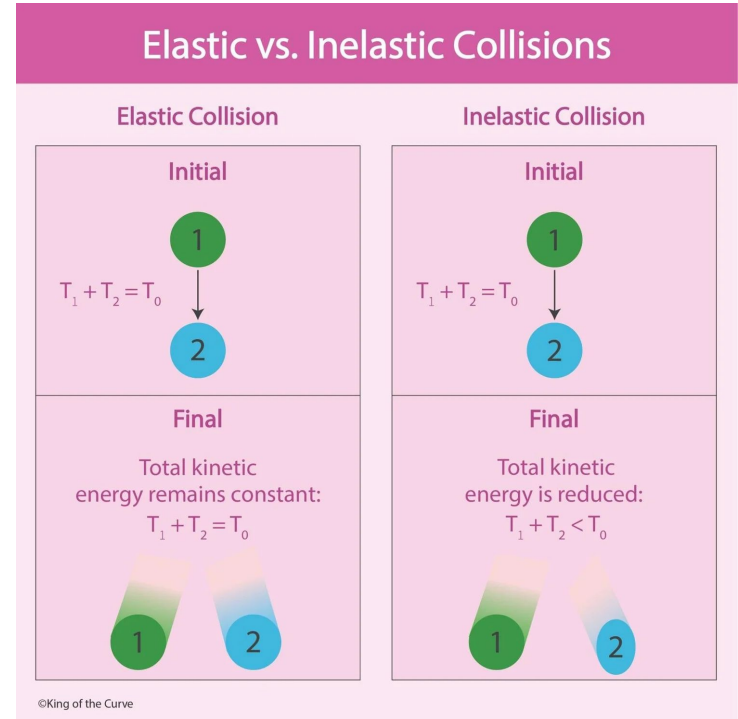
Handwritten equations on a black background:

$$\vec{p} = m\vec{v}$$
$$\vec{F}_{net} = m\vec{a} = \vec{F}_{net} = m \left(\frac{\Delta \vec{v}}{\Delta t} \right)$$
$$\vec{F}_{net} = m \left(\frac{\Delta \vec{p}}{m \cdot \Delta t} \right)$$
$$\vec{F}_{net} = \frac{\Delta \vec{p}}{\Delta t}$$



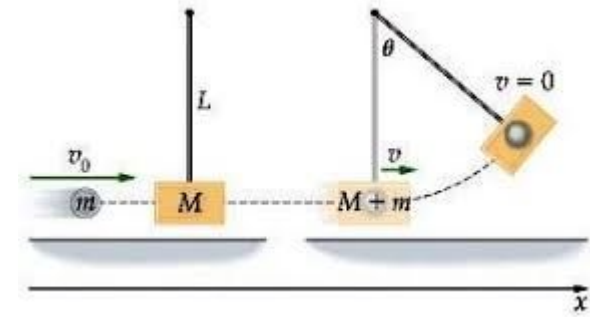
Collisions: Basic Definitions

- Law of conservation of momentum → internal forces only act within a system so momentum is conserved in collisions if friction is negligible
 - $p_0 = p_f$
- Collisions (specifically: linear collisions) → conservation of momentum
 - 2+ objects collide with each other and the motion of at least 1 changes (typically 2)
 - 2 types of collisions:
 - Elastic collisions:
 - Kinetic energy is conserved
 - Objects NEVER stick to each other
 - $m_1 v_{1,i} + m_2 v_{2,i} = m_1 v_{1,f} + m_2 v_{2,f}$
 - Inelastic collisions:
 - Kinetic energy is NOT conserved (lost as heat, sound, etc)
 - Objects MAY stick to each other
 - $m_1 v_{1,i} + m_2 v_{2,i} = m_1 v_{1,f} + m_2 v_{2,f}$ OR $(m_1 + m_2) v_f$

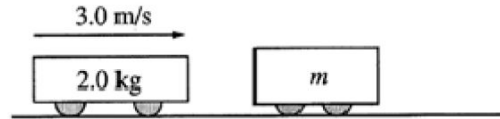


Collision Scenarios To Know

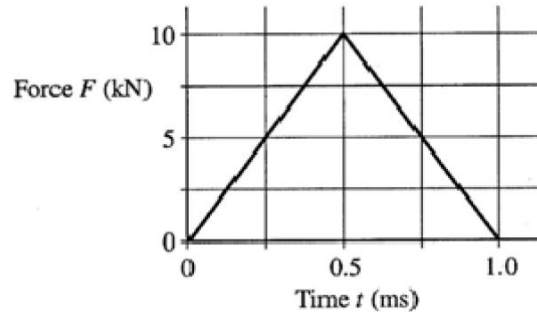
- Ballistic Pendulum
 - Some object collides with a mass on a pendulum at rest (typically a bullet)
 - May either stick or pass through the pendulum
- Simply apply conservation of momentum
 - $p_0 = p_f$



Practice Problem



2002B1B. A 2.0 kg frictionless cart is moving at a constant speed of 3.0 m/s to the right on a horizontal surface, as shown above, when it collides with a second cart of undetermined mass m that is initially at rest. The force F of the collision as a function of time t is shown in the graph below, where $t = 0$ is the instant of initial contact. As a result of the collision, the second cart acquires a speed of 1.6 m/s to the right. Assume that friction is negligible before, during, and after the collision.



- (a) Calculate the magnitude and direction of the velocity of the 2.0 kg cart after the collision.
- (b) Calculate the mass m of the second cart.

Solution

a) $J = m\Delta v = \text{area under } F \text{ vs } t \text{ graph}$
look at units properly

$$\frac{1}{2} (0.001)(10000) = 5$$

USE -5 to account for the sign:

$$-5 = m\Delta v$$

$$-5 = 2(v_f - v_0)$$

$$-5 = 2(v_f - 3) \Rightarrow v_f = 0.5 \text{ m/s}$$

b) $\Delta p = 0$

$$(2)(3) + (m)(0) = (2)(0.5) + (m)(1.6)$$

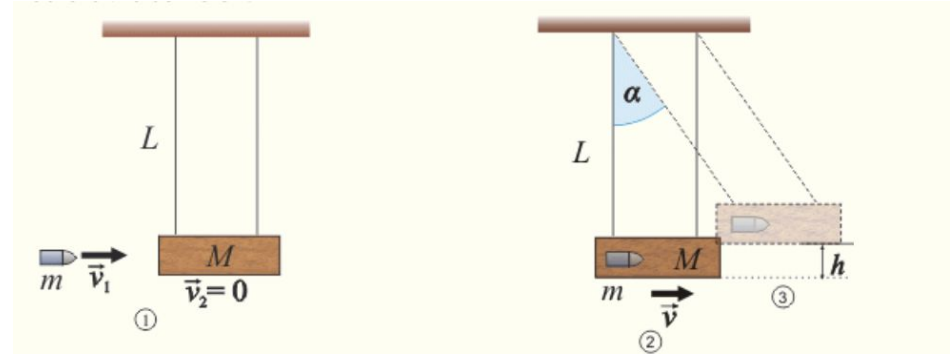
$$6 = 1 + 1.6m$$

$$1.6m = 5$$

$$m = \frac{5}{1.6} = \boxed{3.125 \text{ kg}}$$

Practice Problem

1. A bullet with mass m hits a ballistic pendulum with length L and mass M and lodges in it. When the bullet hits the pendulum it swings up from the equilibrium position and reaches an angle α at its maximum. Determine the bullet's velocity.



Solution

$$\Delta p = 0$$

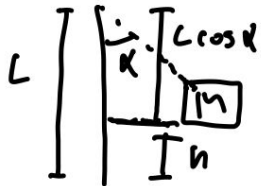
$$p_o = p_f$$

$$(m)(v_i) + (M)(0) = (m+M)v_f$$

$$mv_i = (m+M)v_f$$

$$\Rightarrow v_f = \frac{m}{m+M} v_i$$

at max height



$$h = L - L \cos \alpha = L(1 - \cos \alpha)$$

$$KE = PE$$

$$\frac{1}{2}(m+M)v_f^2 = (m+M)gh$$

$$\frac{1}{2}\left(\frac{m}{m+M}v_i\right)^2 = gL(1 - \cos \alpha)$$

$$\left(\frac{m}{m+M}\right)^2 v_i^2 = 2gL(1 - \cos \alpha)$$

$$v_i^2 = \left(\frac{m+M}{m}\right)^2 \cdot 2gL(1 - \cos \alpha)$$

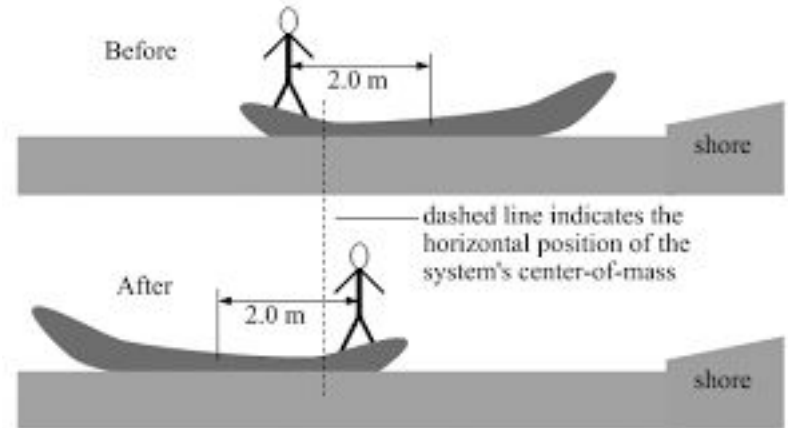
$$v_i = \left(\frac{m+M}{m}\right) \sqrt{2gL(1 - \cos \alpha)}$$

Center of Mass & The Boat Problem

- Coordinate which represents essentially where the system's entire mass can be concentrated

$$x_{cm} = \frac{\sum_{i=1}^N m_i x_i}{M} \quad y_{cm} = \frac{\sum_{i=1}^N m_i y_i}{M}$$

- Extremely useful for “The Boat Problem”
 - Main concept to remember: When ONLY internal forces act within a system, the center of mass does not shift
 - Calculate the initial center of mass, and solve an equation keeping center of mass invariant to find the new positions within the system



Practice Problem

 Two people, one of mass 85 kg and the other of mass 55 kg, sit in a rowboat of mass 58 kg. With the boat initially at rest, the two people, who have been sitting at opposite ends of the boat, 3.0 m apart from each other, now exchange seats. How far and in what direction will the boat move?

Solution

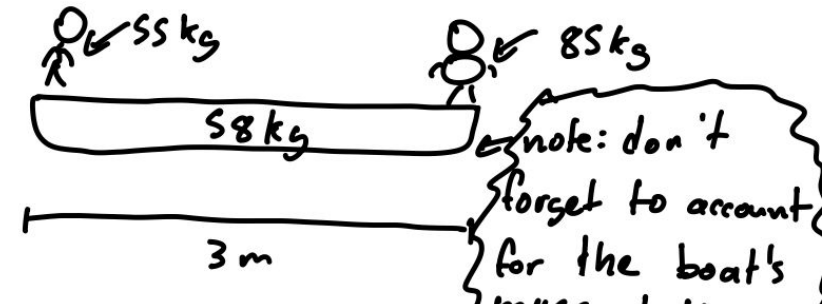


Diagram showing a boat of mass 58 kg with two people on it. The person on the left has a mass of 55 kg, and the person on the right has a mass of 85 kg. The distance between the two people is 3 m. A note indicates: "note: don't forget to account for the boat's mass at the center".

$$CM = \frac{\sum m_i x_i}{\sum m_i} = \frac{55 \cdot 0 + 58 \cdot 1.5 + 85 \cdot 3}{55 + 58 + 85}$$

$$= 1.727$$

CM stays the same $\rightarrow$ no ext. forces

note: boat moves right by d

$$1.727 = \frac{55(3+d) + 58(1.5+d) + 85(d)}{55 + 58 + 85}$$

$d = 0.45 \text{ m}$ $\rightarrow$ -0.45 m is also possible if people start other way